

A Second Look at Classical Gravitational Light Deflection: The Archimedean Energy Contribution in the Hypergeometrical Universe

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Abstract

The classical calculation of the gravitational deflection of light by the Sun is usually said to fail because it gives only half of the observed deflection. This paper revisits that conclusion within the Hypergeometrical Universe (HU) model. In HU, our three-dimensional universe is the surface of a four-dimensional hypersphere expanding at the speed of light, and particles surf the hyperspherical metric wave associated with this expansion. A particle at rest in three-dimensional space is therefore not at rest in four dimensions: it moves radially at c . Applying the ordinary kinetic-energy expression to this motion gives $mc^2/2$, apparently in conflict with the familiar rest energy mc^2 . HU resolves this through an Archimedean displacement picture in which a particle surfing the metric wave displaces an equal mass of that wave. The particle and the displaced metric contribution each carry $mc^2/2$, giving the total energy mc^2 , while a gravitational interaction within the three-dimensional hypersurface involves the particle's $mc^2/2$ share. Using this energy scale in the classical light-deflection calculation gives a deflection angle $4GM/(bc^2)$, or approximately 1.75 arcseconds for light grazing the Sun. Within HU, the classical trajectory calculation therefore does not fail; the historical factor-of-two discrepancy arises from using the total mc^2 energy where the three-dimensional interaction involves only one of the two equal energy contributions.

Keywords: gravitational light deflection; gravitational lensing; Hypergeometrical Universe; Inner Dilation Layer; photon momentum; classical mechanics

I. Introduction and the historical factor-of-two problem

The deflection of starlight by the Sun is one of the classic tests of gravitation [1-4]. A classical corpuscular calculation, using an inverse-square gravitational force, gives a deflection of

$$\delta_{\text{classical}} = \frac{2GM}{bc^2} \quad (1)$$

for a particle passing a mass M with impact parameter b and speed c . Einstein obtained the same result in 1911 [2], before General Relativity. The observed deflection and the later General Relativistic result are twice as large [3,4],

$$\delta_{\text{obs}} = \frac{4GM}{bc^2}. \quad (2)$$

This difference is usually described by saying that classical mechanics fails by a factor of two.

The argument developed here is that the failure is not in the classical mechanics. It is in the representation of the photon used in applying it. The classical calculation represents light as an ordinary particle traveling at the speed of light and then applies the corresponding particle energy and momentum bookkeeping. Within the Hypergeometrical Universe (HU) [5-11], that description is incomplete.

In HU, matter exists on the three-dimensional surface of a four-dimensional hypersphere and surfs the hyperspherical metric wave associated with its expansion. A particle at rest with respect to the three-dimensional hypersurface is therefore still moving radially through the four-dimensional description at speed c . Applying the ordinary kinetic-energy expression to one such moving mass gives

$$\frac{1}{2}mc^2. \quad (3)$$

The familiar total energy mc^2 arises because the particle does not move independently. In the HU picture it surfs the Inner Dilation Layer (IDL) and displaces an equal mass in that metric wave. The two contributions are therefore

$$mc^2 = \frac{1}{2}mc^2 + \frac{1}{2}mc^2. \quad (4)$$

This distinction becomes important in gravitational light deflection. The total HU energy belongs to the combined particle-plus-displaced-IDL system, whereas a gravitational interaction acting within the three-dimensional hypersurface involves the particle's $mc^2/2$ contribution. The classical deflection integral itself is unchanged. What changes is the physical interpretation of the photon to which that calculation is applied.

Using the $mc^2/2$ energy available to the three-dimensional interaction in the fractional momentum or energy change doubles the conventional corpuscular result and gives

$$\delta = \frac{4GM}{bc^2}. \quad (5)$$

The purpose of this paper is therefore not to modify the classical force law or the classical trajectory calculation. It is to show that, within HU, the historical factor-of-two discrepancy originates in treating the photon as an isolated particle moving at c , rather than as a disturbance traveling on and interacting through the four-dimensional hyperspherical metric wave.

II. Classical deflection geometry

Consider a photon passing a spherical body of mass M . To apply the classical gravitational calculation, assign the photon a displacement mass m .

This is only a dynamical bookkeeping parameter and should not be interpreted as a photon rest mass. The gravitational force is

$$F(r) = \frac{GMm}{r^2}. \quad (6)$$

Take the unperturbed photon trajectory to be a straight line along the x -direction, with impact parameter b . At position x , the photon is a distance

$$r = \sqrt{x^2 + b^2} \quad (7)$$

from the center of the gravitating body.

The gravitational force points toward M . Only the component perpendicular to the photon trajectory contributes directly to the deflection. From the geometry of Figure 1,

$$\sin \chi = \frac{b}{r} = \frac{b}{\sqrt{x^2 + b^2}}, \quad (8)$$

and therefore

$$F_{\perp} = F(r) \sin \chi = \frac{GMm}{x^2 + b^2} \frac{b}{\sqrt{x^2 + b^2}} = \frac{GMm b}{(x^2 + b^2)^{3/2}}. \quad (9)$$

Here χ denotes only the instantaneous geometrical angle used to resolve the gravitational force. The final deflection angle will be denoted by δ . The angle used elsewhere in HU to describe radial and tangential motion in the four-dimensional hypersphere is a different quantity and will be given a separate symbol.

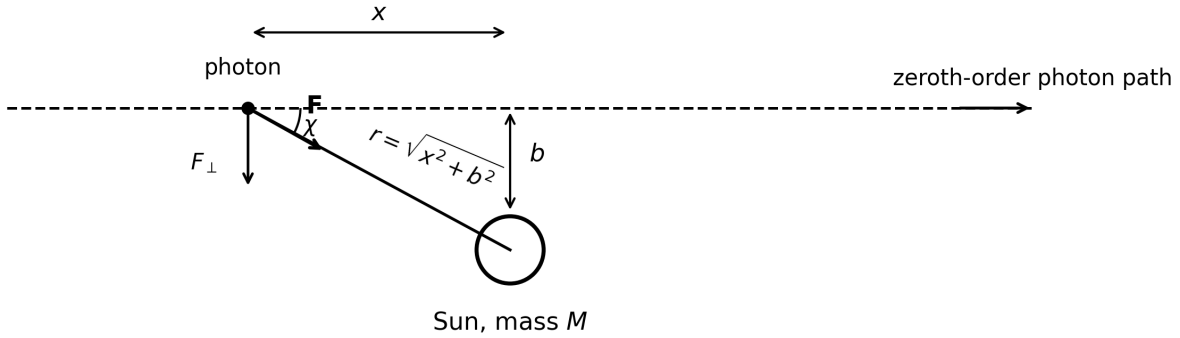


Figure 1. Geometry of the classical deflection calculation. The zeroth-order photon trajectory passes the gravitating mass M at impact parameter b . The coordinate x measures distance along the unperturbed trajectory, $r = \sqrt{x^2 + b^2}$ is the instantaneous distance from M to the photon, and χ resolves the radial gravitational force into the component $F_{\perp}$ perpendicular to the photon path.

III. Energy and momentum of a photon in a gravitational flyby

For a photon,

$$E = pc = h\nu, \quad (10)$$

where $p = |\mathbf{p}|$. A gravitational interaction therefore changes the photon energy through its momentum and frequency, not through a change in its propagation speed.

During the flyby, both the magnitude and the direction of the momentum can change. We may write

$$\mathbf{p}(t) = p(t) \hat{\mathbf{n}}(t), \quad (11)$$

where $p(t)$ determines the photon energy and $\hat{\mathbf{n}}(t)$ gives its direction of propagation.

As the photon falls into the gravitational potential, it gains energy and is blueshifted. As it climbs out, that energy is returned and the photon is redshifted back to its original asymptotic frequency. The change in direction, however, does not cancel.

Using the straight-line trajectory introduced in Section 2,

$$r = \sqrt{x^2 + b^2}, \quad (12)$$

the component of the gravitational force along the unperturbed direction of motion is

$$F_x = -\frac{GMm x}{(x^2 + b^2)^{3/2}}. \quad (13)$$

For the incoming half of the trajectory, $x < 0$, so $F_x > 0$ and the gravitational field does positive work on the photon:

$$\Delta E_{\text{in}} = \int_{-\infty}^0 F_x dx = \frac{GMm}{b}. \quad (14)$$

At closest approach the photon has therefore gained the energy

$$\Delta E_{\text{max}} = \frac{GMm}{b}. \quad (15)$$

On the outgoing half, $x > 0$, and the force component along the trajectory reverses sign. The photon returns the energy it gained on the way in:

$$\Delta E_{\text{out}} = \int_0^{+\infty} F_x dx = -\frac{GMm}{b}. \quad (16)$$

Thus, when the initial and final gravitational potentials are the same,

$$\Delta E_{\text{in}} + \Delta E_{\text{out}} = 0. \quad (17)$$

The transverse force behaves differently. Its direction toward the gravitating mass is the same on both halves of the trajectory, so the transverse momentum changes add rather than cancel. To leading order $dt = dx/c$, and therefore

$$\begin{aligned} \Delta p_{\perp} &= \int_{-\infty}^{+\infty} F_{\perp} dt \\ &= \frac{1}{c} \int_{-\infty}^{+\infty} \frac{GMm b}{(x^2 + b^2)^{3/2}} dx \\ &= \frac{2GMm}{bc}. \end{aligned} \quad (18)$$

The photon therefore leaves the potential well with the same asymptotic energy it had on entering, but with a different momentum direction.

Because $E = pc$, it is convenient to multiply the transverse momentum change by c :

$$c \Delta p_{\perp} = \frac{2GMm}{b}. \quad (19)$$

This quantity has units of energy and allows the deflection calculation to be written in the same energy form that motivated the original analysis,

$$\frac{\Delta p_{\perp}}{p} = \frac{c \Delta p_{\perp}}{E}. \quad (20)$$

The distinction is straightforward. The photon gains the energy GMm/b while falling into the potential and returns it while climbing out. The transverse momentum accumulated during the two halves of the flyby does not cancel, and it is this remaining change in momentum direction that produces the observed deflection.

IV. The HU radial-energy puzzle

The $mc^2/2$ energy scale used here was not introduced to solve the light-deflection problem. It follows from an earlier question in the HU description of matter [7].

In HU, ordinary three-dimensional space is the hypersurface of a four-dimensional hypersphere whose radius expands at the speed of light. Matter is carried by, or “surfs,” the hyperspherical metric wave associated with this expansion. A body that is at rest in three-dimensional space is therefore not at rest in the four-dimensional description. Its motion is entirely radial, at speed c .

Consider a body of rest mass m_0 . $K = \frac{1}{2} m_0 v^2$, if the ordinary kinetic-energy expression is applied to a single mass contribution moving radially at c ,

then

$$K_{\text{radial}} = \frac{1}{2} m_0 c^2. \quad (21)$$

The familiar rest energy associated with the same body is

$$E_0 = m_0 c^2. \quad (22)$$

Equations (21) and (22) differ by exactly a factor of two.

There is no contradiction with standard relativity. In Special Relativity, $m_0 c^2$ is the rest energy and is not obtained by inserting $v = c$ into the nonrelativistic expression $m_0 v^2/2$.

HU asks a different question. If radial motion through the fourth spatial direction and tangential motion within the three-dimensional hypersurface are manifestations of the same underlying motion, why should the kinetic-energy bookkeeping be different for the two?

Within HU, Eq. (21) is therefore taken seriously. The factor of two suggests that the total $m_0 c^2$ energy does not belong to a single $m_0 c^2$ kinetic contribution. Instead, a particle surfing the hyperspherical

metric wave carries two equal kinetic-energy contributions. One belongs to the particle itself; the other belongs to the equal mass displaced in the Inner Dilation Layer.

The next section develops that Archimedean interpretation.

V. Archimedean/IDL of the second $mc^2/2$ contribution

In HU, a material particle does not move independently through the four-dimensional hyperspherical structure. It surfs the Inner Dilation Layer (IDL), the hyperspherical metric wave associated with the expansion of the Universe.

The Archimedean idea enters because the particle's motion through this layer is accompanied by a displacement of the layer itself. HU assigns to that displaced portion of the IDL the same mass-equivalent as the material particle,

$$m_{\text{IDL}} = m. \quad (23)$$

This is the origin of the two mass contributions in the HU description.

Applying the same kinetic-energy expression to each contribution at the limiting speed c gives

$$K_{\text{particle}} = \frac{1}{2}mc^2, \quad K_{\text{IDL}} = \frac{1}{2}m_{\text{IDL}}c^2 = \frac{1}{2}mc^2. \quad (24)$$

The total energy of the particle together with the displaced IDL contribution is therefore

$$\begin{aligned} E_{\text{total}} &= K_{\text{particle}} + K_{\text{IDL}} \\ &= \frac{1}{2}mc^2 + \frac{1}{2}mc^2 \\ &= mc^2. \end{aligned} \quad (25)$$

Thus the familiar mc^2 energy scale is recovered without assigning mc^2 to a single kinetic-energy contribution. Each of the two participating mass contributions carries the same classical kinetic-energy share,

$$\frac{1}{2}mc^2. \quad (26)$$

This is the point of the Archimedean construction. A body that is stationary in three-dimensional space still moves radially at c with the expanding hypersphere. Its material component contributes $mc^2/2$, while the equal mass-equivalent displaced in the IDL contributes the second $mc^2/2$. Together they give the total mc^2 .

The use of the word *Archimedean* here describes the displacement picture, not the ordinary fluid-mechanical theorem. In a conventional fluid, the displaced mass depends on the density of the fluid and the displaced volume. The equality in Eq. (23),

$$m_{\text{IDL}} = m, \quad (27)$$

is a property proposed by the HU matter/IDL construction. It is this equality that produces the two equal energy contributions.

VI. Why a three-dimensional interaction uses $mc^2/2$

The second HU proposition concerns which part of the total particle-plus-IDL energy budget participates in motion confined to the ordinary three-dimensional hypersurface. In the Archimedean/IDL construction developed above, the total energy scale is

$$E_{\text{total}} = \frac{1}{2}mc^2 + \frac{1}{2}mc^2 = mc^2, \quad (28)$$

where one share is associated with the material component and the second with the displaced IDL contribution.

The geometrical distinction between these two contributions is essential. The displacement of the IDL is radial in the embedding space, and the corresponding Archimedean, or buoyancy-like, force is normal to the three-dimensional hypersurface. By contrast, ordinary three-dimensional motion is tangent to that hypersurface. Thus, for an infinitesimal tangential displacement $d\mathbf{r}_{\parallel}$,

$$\mathbf{F}_{\text{IDL}} \perp d\mathbf{r}_{\parallel}, \quad (29)$$

and consequently

$$dW_{\text{IDL}} = \mathbf{F}_{\text{IDL}} \cdot d\mathbf{r}_{\parallel} = 0. \quad (30)$$

The displaced-IDL contribution therefore participates in the radial energy accounting required to recover the total mc^2 scale, but it performs no work on motion tangent to the hypersurface. A force producing ordinary three-dimensional motion couples only to the material share of the energy budget.

The maximum kinetic-energy scale available to such a three-dimensional interaction is therefore

$$E_{3D} = \frac{1}{2}mc^2 \quad (31)$$

The distinction is not that the other half of the total energy is unreal or absent. On the contrary, HU assigns that second half to the displaced IDL contribution, and both shares are required to recover

$$E_{\text{total}} = mc^2. \quad (32)$$

Rather, the distinction concerns the direction in which each contribution can perform work. The IDL contribution acts radially, whereas gravitational deflection occurs within the tangent three-dimensional hypersurface. Since a radial force performs no work on an infinitesimal tangential displacement, the IDL share does not enter the tangential interaction-energy denominator.

For gravitational light deflection, Eq. (31) is therefore the substantive nonstandard input of the HU calculation.

The same point can be expressed in momentum language without changing the energy-based interpretation. For a small deflection angle δ ,

$$\delta \simeq \frac{\Delta p_{\perp}}{p}. \quad (33)$$

Using

$$E = pc \quad (34)$$

and the previously defined energy-dimension deflection increment

$$\Delta E_{\text{defl}} \equiv c \Delta p_{\perp}, \quad (35)$$

one obtains

$$\delta = \frac{\Delta p_{\perp}}{p} = \frac{c \Delta p_{\perp}}{pc} = \frac{\Delta E_{\text{defl}}}{E} \quad (36)$$

Equation (36) is an identity once $E = pc$ and

$$\Delta E_{\text{defl}} = c \Delta p_{\perp} \quad (37)$$

have been defined. The physics therefore lies entirely in the choice of the energy scale E .

If the denominator is taken to be the full

$$E = mc^2, \quad (38)$$

the classical calculation gives the historical half-value of the observed solar deflection. Within HU, however, the force responsible for the deflection acts tangentially to the hypersurface, while the displaced-IDL contribution is associated with the normal direction and performs no tangential work. The relevant interaction-energy scale is consequently

$$E_{3D} = \frac{1}{2} mc^2, \quad (39)$$

and the resulting deflection coefficient is doubled.

This geometrical separation between radial IDL energy and tangential interaction energy is the central HU reason for using $mc^2/2$ rather than mc^2 in the classical light-deflection calculation.

VII. Light-deflection calculation

The small-angle approximation also clarifies the physical meaning of the energy and momentum changes during the encounter. Two gravitational effects occur simultaneously and should be distinguished. First, as the photon moves into the Sun's gravitational potential well, it gains energy and is gravitationally blueshifted. For the undeflected trajectory,

$$\Delta E_{\gamma}(x) = \frac{GMm}{\sqrt{x^2 + b^2}}, \quad (40)$$

relative to its energy at infinity. At closest approach the increase is

$$\Delta E_{\gamma, \max} = \frac{GMm}{b}. \quad (41)$$

On the outgoing trajectory this process is reversed. The photon is redshifted as it climbs out of the potential well and returns asymptotically to its original energy. Thus this scalar energy change is reversible:

$$\Delta E_{\gamma}(-\infty) = \Delta E_{\gamma}(+\infty) = 0, \quad (42)$$

when measured relative to the asymptotic photon energy. The photon therefore gains energy on the way into the potential well and gives that energy back on the way out.

At the same time, the gravitational force produces a transverse impulse. In contrast with the scalar energy change, the transverse contributions from the incoming and outgoing portions of the trajectory have the same sign and therefore accumulate. Equation (19) gives

$$\Delta p_{\perp} = \frac{2GMm}{bc}. \quad (43)$$

It is useful to express this accumulated transverse impulse in energy units by defining

$$\Delta E_{\text{defl}} \equiv c \Delta p_{\perp} = \frac{2GMm}{b}. \quad (44)$$

The quantity ΔE_{defl} should not be interpreted as the net scalar energy gained by the photon. It is the accumulated transverse impulse expressed in energy units. The actual scalar photon energy changes because of its changing position in the gravitational potential: it increases during infall and decreases by the same amount during escape. The transverse momentum change, however, does not reverse and survives as the final change in propagation direction.

This distinction makes the meaning of Eq. (36) explicit:

$$\delta \simeq \frac{\Delta p_{\perp}}{p} = \frac{c \Delta p_{\perp}}{E} = \frac{\Delta E_{\text{defl}}}{E}. \quad (45)$$

The ratio $\Delta p_{\perp}/p$ is extremely small for a solar grazing ray, so the undeflected path may consistently be used in evaluating the force integral. No nonrelativistic propagation speed is assigned to the photon. Its local speed remains c ; its energy and momentum magnitude vary through the gravitational blueshift and redshift, while the transverse component of momentum accumulates to produce the permanent angular deflection.

VIII. Solar numerical evaluation

For a grazing ray, take $b = R_{\odot}$ and use [17,18]

$$\begin{aligned} G &= 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ M_{\odot} &= 1.9885 \times 10^{30} \text{ kg}, \\ R_{\odot} &= 6.96 \times 10^8 \text{ m}, \end{aligned} \quad (46)$$

and

$$c = 2.99792458 \times 10^8 \text{ m s}^{-1}. \quad (47)$$

Equation (45) then gives

$$\delta_{\text{HU}} = \frac{4GM_{\odot}}{R_{\odot} c^2} = 8.4867 \times 10^{-6} \text{ rad}. \quad (48)$$

Using 206264.806 arcseconds per radian,

$$\delta_{\text{HU}} = 1.7505 \text{ arcsec}. \quad (49)$$

Thus the classical transverse-force calculation, when combined with the HU three-dimensional interaction-energy scale $E_{3\text{D}} = mc^2/2$, reproduces the standard leading solar-deflection coefficient. The numerical agreement has a limited but precise significance: the solar deflection coefficient by itself cannot distinguish between theories [13-15] that predict the same leading expression,

$$\delta = \frac{4GM}{bc^2}. \quad (50)$$

That last formulation is stronger scientifically because it says exactly what happened: **classical trajectory mechanics supplied the impulse; HU supplied the energy denominator**. It also avoids implying that the force law itself was altered.

IX. Comparison with General Relativity

General Relativity predicts the same leading weak-field deflection,

$$\delta_{\text{GR}} = \frac{4GM}{bc^2}. \quad (51)$$

The relativistic result is sometimes described heuristically as consisting of a Newtonian contribution plus an equal contribution associated with spatial curvature. Although useful for intuition, that separation depends on how the relativistic calculation is formulated and should not be regarded as a unique physical decomposition.

The comparison is particularly transparent in the parametrized post-Newtonian form [12,16],

$$\delta_{\text{PPN}} = 2(1 + \gamma_{\text{PPN}}) \frac{GM}{bc^2}, \quad (52)$$

where General Relativity has

$$\gamma_{\text{PPN}} = 1. \quad (53)$$

Equation (52) then gives

$$\delta_{\text{GR}} = \frac{4GM}{bc^2}. \quad (54)$$

Setting

$$\gamma_{\text{PPN}} = 0 \tag{55}$$

gives

$$\delta = \frac{2GM}{bc^2}, \tag{56}$$

which is numerically the same coefficient obtained in the historical Newtonian/Soldner corpuscular calculation [1].

HU reaches the observed coefficient by a different route. The classical gravitational force and the transverse-force integration are left unchanged. The factor of two enters when the transverse impulse is converted into a deflection angle. In the ordinary corpuscular calculation the energy denominator is

$$E = mc^2, \tag{57}$$

whereas HU assigns a force acting tangentially within the three-dimensional hypersurface the interaction-energy scale

$$E_{3\text{D}} = \frac{1}{2} mc^2. \tag{58}$$

Consequently,

$$\delta_{\text{HU}} = \frac{4GM}{bc^2}, \tag{59}$$

the same leading result obtained in General Relativity.

The agreement at this order does not imply that the underlying mechanisms are the same. In General Relativity, the deflection follows from photon propagation in curved spacetime. In HU, it follows from the classical transverse gravitational impulse together with the HU particle-plus-IDL energy partition and the fact that the displaced-IDL contribution performs no work on tangential motion.

The solar grazing-ray measurement therefore establishes the leading coefficient

$$\frac{4GM}{bc^2}, \tag{60}$$

but that coefficient alone does not determine which physical mechanism produced it.

X. Assumptions, scope, and falsifiability

The ingredients entering the calculation are summarized in Table 1. The classical force calculation and photon energy-momentum relation are distinct from the HU-specific assumptions. The factor of two comes from the latter, not from a change in the inverse-square force or in the transverse-force integral.

Table 1. Logical status of the ingredients used in the derivation.

Ingredient	Status in this paper	Role in the result
$F = \frac{GMm}{r^2}$	Classical bookkeeping assumption using a photon displacement mass m	Defines the force and transverse-impulse calculation
$E = pc = h\nu$	Standard photon energy-momentum relation	Connects photon energy, momentum magnitude, and frequency
Straight zeroth-order path	Leading small-angle approximation	Allows the force integral to be evaluated analytically
$m_{\text{IDL}} = m$	HU-specific proposition	Produces equal material and displaced-IDL mass contributions
E_{total} $= \frac{1}{2}mc^2 + \frac{1}{2}mc^2$ $= mc^2$	HU energy partition	Reconciles the radial $mc^2/2$ contribution with the total mc^2 energy scale
$\mathbf{F}_{\text{IDL}} \cdot d\mathbf{r}_{\parallel} = 0$	Geometrical consequence of the HU construction	Prevents the radial IDL contribution from doing work on tangential motion
$E_{3\text{D}} = \frac{1}{2}mc^2$	HU three-dimensional interaction-energy scale	Supplies the factor of two in the deflection denominator

The derivation therefore stands or falls with the HU-specific ingredients. If the IDL construction does not support

$$m_{\text{IDL}} = m, \quad (61)$$

then the two equal $mc^2/2$ contributions do not follow. Likewise, if the displaced IDL contribution can perform work on motion tangent to the hypersurface, then the use of

$$E_{3\text{D}} = \frac{1}{2}mc^2 \quad (62)$$

in the light-deflection calculation is not justified. In either case, the factor-of-two argument fails.

The reverse implication does not follow. Recovering the observed solar coefficient,

$$\delta = \frac{4GM}{bc^2}, \quad (63)$$

does not by itself establish the IDL construction or the energy partition [12-15], because that same leading coefficient is already obtained in General Relativity. The value of the present calculation is therefore not that it provides a new numerical prediction for solar deflection, but that it shows how the classical half-value is removed within the HU description without changing the classical transverse-force calculation.

At leading order,

$$\delta_{\text{HU}} = \delta_{\text{GR}} = \frac{4GM}{bc^2}. \quad (64)$$

Solar light deflection at this order therefore cannot distinguish the two mechanisms. A discriminating test must involve an observable for which HU and General Relativity make different predictions while the same HU assumptions are retained. Such a test could arise from higher-order light deflection or from another consequence of the particle-plus-IDL dynamics, but developing those predictions lies beyond the present calculation.

XI. Discussion

The calculation separates three effects that are easily confused in gravitational light deflection.

First, the photon changes energy as it moves through the gravitational potential. On the incoming trajectory it falls deeper into the potential well, its frequency increases, and it is blueshifted. On the outgoing trajectory the process reverses. In a static symmetric field the photon eventually returns to the same asymptotic energy with which it entered:

$$\Delta E_\gamma(-\infty) = \Delta E_\gamma(+\infty) = 0, \quad (65)$$

when the energy change is measured relative to its value at infinity.

Second, returning this scalar energy does not undo the deflection. The transverse gravitational force points toward the gravitating body on both halves of the trajectory. Its contributions therefore add,

$$\Delta p_\perp = \frac{2GMm}{bc}, \quad (66)$$

and the photon leaves the encounter with essentially the same asymptotic energy but a different direction of propagation.

These two effects can be represented without changing from an energy description to a different physical picture. For a photon,

$$E = pc, \quad (67)$$

so the deflection may be written

$$\delta \simeq \frac{\Delta p_\perp}{p} = \frac{c \Delta p_\perp}{E}. \quad (68)$$

Defining

$$\Delta E_{\text{defl}} \equiv c \Delta p_\perp \quad (69)$$

is simply a convenient way of expressing the accumulated transverse impulse in energy units. It is not the scalar energy gained by the photon while falling into the potential well. The latter is gained and returned; the transverse momentum change remains.

The third question is the origin of the historical factor of two. Nothing in the transverse-force integral produces it. The classical calculation gives

$$c \Delta p_{\perp} = \frac{2GMm}{b}. \quad (70)$$

The difference appears only when this quantity is divided by the energy scale governing the three-dimensional response. Using

$$E = mc^2 \quad (71)$$

gives the Newtonian/Soldner result,

$$\delta = \frac{2GM}{bc^2}. \quad (72)$$

HU instead gives the material three-dimensional share

$$E_{3D} = \frac{1}{2} mc^2, \quad (73)$$

and therefore

$$\delta = \frac{2GMm/b}{mc^2/2} = \frac{4GM}{bc^2}. \quad (74)$$

The origin of $mc^2/2$ is independent of the lensing integral. In HU, a material particle surfs the Inner Dilation Layer and displaces an equal mass-equivalent of that layer. The two contributions give

$$E_{\text{total}} = \frac{1}{2} mc^2 + \frac{1}{2} mc^2 = mc^2. \quad (75)$$

The displaced-IDL contribution belongs to the radial motion of the particle-IDL system. Its associated Archimedean force is normal to the three-dimensional hypersurface and therefore does no work on a tangential displacement,

$$\mathbf{F}_{\text{IDL}} \cdot d\mathbf{r}_{\parallel} = 0. \quad (76)$$

Consequently, the IDL contribution remains part of the total mc^2 energy while the energy scale governing tangential three-dimensional motion is the material share $mc^2/2$.

This is why the $mc^2/2$ term should not be interpreted as the standard relativistic kinetic energy of a massive particle moving at c . Standard relativity does not make that identification. Massive-particle energy is governed by the Lorentz factor, while photons satisfy

$$E = pc \quad (77)$$

and possess no rest frame. The $mc^2/2$ appearing here has a different origin: it is one of the two kinetic-energy contributions in the HU particle-IDL construction.

The solar-deflection calculation is therefore a consequence of that broader HU picture rather than a special correction introduced for gravitational lensing. The classical force integration supplies the transverse impulse; the particle-IDL structure determines which energy share responds to motion within the three-dimensional hypersurface. Together they give the observed leading coefficient

$$\delta_{\text{HU}} = \frac{4GM}{bc^2}. \quad (78)$$

XII. Conclusion

The historical classical calculation of solar light deflection gives

$$\delta_{\text{classical}} = \frac{2GM}{bc^2}, \quad (79)$$

one half of the observed weak-field deflection and of the leading result of General Relativity. The argument developed here suggests that, within the Hypergeometrical Universe framework, the missing factor of two does not originate in the classical force integration. The inverse-square force acting along the zeroth-order trajectory gives the transverse impulse

$$\Delta p_{\perp} = \frac{2GMm}{bc}, \quad (80)$$

exactly as in the classical calculation. The difference enters through the HU description of matter and of the Inner Dilation Layer.

The IDL is the expanding metric wave of the HU Cosmogenesis model. It is not the source of the material particles. Matter originates instead in the Outermost Contraction Layer (OCL). In the initial Cosmogenesis configuration, the contraction energy of the OCL is balanced by the dilation energy of the IDL. The IDL carries the negative-energy side of this balance, so that

$$E_{\text{OCL}} + E_{\text{IDL}} = 0. \quad (81)$$

If the contraction and dilation were simply allowed to recombine, their opposite deformations would cancel, space would return to its undeformed state, and another cosmological cycle could begin.

The present Universe arises because the expanding IDL pushes against and fragments the OCL. Those fragments become the material particles. The fragmentation also prevents the contraction and dilation sectors from immediately recombining. In the HU Cosmogenesis picture, this creates the entropic barrier that allows the fragmented Universe to persist until their eventual recombination.

The connection with the present light-deflection argument is direct. A material particle produced from the fragmented OCL subsequently surfs the IDL wave. Its motion therefore involves both the material particle and a displacement of the IDL.

Because the IDL carries negative energy, displacement of the IDL produces a positive energy contribution. If a displaced region of the undisturbed IDL contains

$$E_{\text{IDL}} < 0, \quad (82)$$

then removing that negative-energy contribution produces

$$\Delta E_{\text{disp}} = -E_{\text{IDL}} > 0. \quad (83)$$

Thus the positive energy associated with the displaced IDL is not in conflict with the negative energy carried by the IDL itself. It follows precisely because what is being displaced is a negative-energy component.

For the equal-displacement relation

$$m_{\text{IDL}} = m, \quad (84)$$

the HU particle-IDL construction gives two equal positive energy contributions,

$$E_{\text{material}} = \frac{1}{2}mc^2, \quad (85)$$

and

$$E_{\text{disp}} = \frac{1}{2}mc^2, \quad (86)$$

so that

$$E_{\text{total}} = E_{\text{material}} + E_{\text{disp}} = \frac{1}{2}mc^2 + \frac{1}{2}mc^2 = mc^2. \quad (87)$$

The two contributions do not, however, play identical roles in three-dimensional motion. The Archimedean force associated with the displaced IDL acts in the radial direction of the hyperspherical construction and is normal to the three-dimensional hypersurface. For a tangential displacement,

$$\mathbf{F}_{\text{IDL}} \cdot d\mathbf{r}_{\parallel} = 0. \quad (88)$$

The displaced-IDL contribution therefore belongs to the complete particle-IDL energy budget but performs no work on motion tangent to the hypersurface. The energy scale governing such a three-dimensional interaction is consequently the material share,

$$E_{3\text{D}} = \frac{1}{2}mc^2. \quad (89)$$

For the photon flyby, the scalar photon energy changes as the photon moves through the gravitational potential. It is blueshifted while falling into the potential well and redshifted as it climbs out, returning to its original asymptotic energy in a static symmetric field. The transverse gravitational impulse behaves differently: its contributions on the incoming and outgoing portions of the trajectory have the same deflecting sign and therefore accumulate.

Writing

$$\Delta E_{\text{defl}} \equiv c \Delta p_{\perp} = \frac{2GMm}{b}, \quad (90)$$

where ΔE_{defl} is the accumulated transverse impulse expressed in energy units rather than the scalar energy acquired in the potential well, the small-angle relation becomes

$$\delta = \frac{\Delta p_{\perp}}{p} = \frac{\Delta E_{\text{defl}}}{E}. \quad (91)$$

Using the HU three-dimensional interaction-energy scale then gives

$$\delta_{\text{HU}} = \frac{2GMm/b}{mc^2/2} = \frac{4GM}{bc^2}. \quad (92)$$

For a ray grazing the Sun this is approximately

$$1.75 \text{ arcsec}. \quad (93)$$

The result therefore does not claim that ordinary Newtonian mechanics alone predicts the observed light deflection. With the conventional mc^2 denominator, the historical corpuscular calculation still gives $2GM/(bc^2)$. The HU claim is instead that the classical transverse-force calculation was not the source of the missing factor of two. Within HU, that factor follows from the physical representation of the particle: matter originates through fragmentation of the OCL, surfs the negative-energy IDL wave, displaces part of that wave with a corresponding positive displacement energy, and responds tangentially through only the material $mc^2/2$ share.

General Relativity and HU consequently give the same leading solar coefficient,

$$\delta = \frac{4GM}{bc^2}, \quad (94)$$

but arrive there through different physical mechanisms. In General Relativity, light deflection follows from propagation in curved spacetime. In HU, it follows from the classical transverse gravitational impulse together with the particle-IDL structure inherited from Cosmogenesis.

The central issue is therefore no longer the elementary solar-deflection integral. It is whether the same HU Cosmogenesis model independently supports the chain

$$\text{OCL fragmentation} \rightarrow \text{matter} \rightarrow \text{particle-IDL coupling} \rightarrow m_{\text{IDL}} = m \rightarrow E_{3\text{D}} = \frac{1}{2}mc^2 \quad (95)$$

while simultaneously maintaining the zero-energy balance between OCL contraction and IDL dilation. If that structure is correct, the full solar light-deflection coefficient follows without altering the classical transverse-force calculation.

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